2026

Analyst Lab Africa

Experience Lab Internship

FinTrust Digital Bank: Final Testing and Validation

Data Analytics Track

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Week 4: FINAL ANALYSIS REPORT

1. Project Overview

Week 4 is the final stage of the FinTrust Digital Bank Experience Lab Project. Week 4 is about testing, validating, Refining, Finalizing and Presenting. This report consolidates the Data Aanalytics track's final output by confirming data quality and calculation accuracy, reviewing the most important SQL analyses, finalizing the Power BI dashboard, and presenting five validated business insights with recommendations. It builds on three independently built analytical layers (Excel, SQL and Python) and one integrated reporting layer Power BI), all connected to the same underlying FinTrust datasets: Customer and Transaction data across Q1 2026(Jan-Mar).

2. Part A: Final Data & Analysis Review

Data Quality Check: All prepared datasets are comfirmed clean. A relationship check between the customer and transaction tables found zero orphan transactions and zero customers without transactions, every one of the 12,000 transactions matches a valid customer, and every one of the 1,500 customers has at least one transaction on record. A well balance data quality integrity on Customer_ID is sound throughout the datasets.

Calculation accuracy: All calculations are confirm accurate and neat. The core KPIs (total customers, total transactions, total value which is split into all-status and successful-only, success rate, and risk-review rate) were independently recomputed in Excel, SQL, and Python, and reconciled against the Power BI dashboard. All four produced identical figures.

SQL results: All SQL outputs are confirm accurate and correct as expected. Twelve business quaries were run againstthe cleaned datasets, using JOINS, GROUP BY, CASE-based bucketing, a window function (NTILE), and a CTE. Every result was internally consistent with the figures produced independently in the other tools. For Example, the SQL-derived successful value of ₦510,807,441 matches the Excel and Python figures exactly.

Python results: All python codes and outputs are confirm accurate and correct. The notebook rebuilds the full customer-level model and RFM segmentation deterministically from the two raw CSV files with no manual intervention required, meaning a rerun from scratch reproduces the same output every time.

KPI definitions: All built KPIs are clearly and consistently defined across all tools;

- **Total Value (all-status):** Sum of Amount_NGN across every transaction regardless of outcome.
- Total Value (successful)**:** Sum of Amount_NGN restricted to Transaction_Status= 'Successful', the figure used for all value-based analysis.
- **Success Rate:** Share of transactions with status successful.
- **Risk Review Rate:** Share of transactions with Risk_Review_Flag= 'Yes'.

Evidence for findings: Every finding in this report cites the specific query, sheet, or notebook cell it was drawn from, and every headline figure has been cross-checked in at least two of the four tools before being written up.

Correction made this week: An earlier draft of the wwek 3 report stated the high-value transaction (above ₦100,000) flag rate as 30.2%. Re-verified directly against the raw transaction data this week, the correct figure is 30.0%. This is documented formally as a validation change in Part E.

3. Part B: Final SQL Analysis

The SQL layer comprises twelve business queries(Q0-Q11) built against the cleaned FinTrust tables, demonstrating JOINs, GROUP BY aggregation, CASE-based classification, a CTE, and a window function (NTILE). The five most important for this final submission are as follows:

Q1 - Transaction status breakdown: 90.47% of the 12,000 transactions succeed; the remaining 9.53% (Failed, Reversed, Pending) account for roughly ₦49.7million in value, which is about 8.9% of the all-status total.

What it reveals: This single query is the foundation of the report's most important correction. It proofs that the headline ₦560.48million figure includes money that never actually settled, and that successful-only value (₦510.81 million) is the number that should anchor every downstream decision.

Q2 - Value by transaction type: Transfers alone generate ₦218.4 million, 42.7% of all successful value from 3,213 transactions, with a mean (₦67,963) running well above the median (₦15,126) in every transaction type.

What it reveals: Transfer is FinTrust's core product by value, and the gap between mean and median across every category is a reminder to describe "typical" transactions using medians, not averages, when the data is this skewed.

Q5 - Risk-review rate by transaction type: Transfer (28.5%) and Cash Withdrawal (25.3%) are flagged for review roughly twice as often as Bill Payment (12.8%) or Card Purchase (13.0%).
What it reveals: Review attention concentrates specifically on transactions where money leaves the customer's control directly, an explainable, sensible pattern rather than a random one, and one that should shape how review capacity is staffed.

Q7 - Risk review rate by transaction amount band: The flag rate holds steady between 16% and 19% for transactions under ₦100,000, then climbs sharply by 25.5% for ₦100,000-25,000, and 40.9% for anything above ₦250,000.
What it reveals: This is the clearest threshold effect in the entire analysis, transaction size is, on its own, a strong driver of review flags, independent of type or channel. It is also the query that confirms the corrected 30.0% high-value flag rate referenced in Part A.

Q11 - Customer value concentration (window function): Using NTILE to rank customers into value deciles, the top 10% (150 customers) generate 27.0% of successful value, the top 30% generate 59.5% and the bottom half generate only 20.0%.
What it reveals: This is the SQL-native confirmation of the concentration pattern found separately in Excel and python, arriving at essentially the same answer through a third, independent method is exactly the kind of triangulation the Week 4 validation standard asks for.

A data-governance finding worth flagging on its own (Q9): Dormat customers average 8.1 transactions each, and Restricted customers average 8.6; both higher than Active customers at 8.0
What it reveals: The account_Status field does not appear to track nreal transaction behaviour. This isn't a business insights so much as a data quality flag, which is worth raising with FinTrust as question about whether the status field is being kept current.

4. Part C: Final Power BI Dashboard

This dashboard connects directly to raw customer and transaction data, not to pre-aggregated Excel output; so that its figures stand as an independent validation of Excel, SQL, and Python layers rather than a restatement of any one of them. It presents six KPI cards (Total Customers, Total Transactions, Total Value, Average Transaction Value, Success Rate, Risk Review Rate), a monthly trend with month-over-month growth, a customer-segment breakdown alongside a Top 10 Customers by Value chart, a Transaction Type x Channel risk matrix, a transaction-status breakdown, and a value-by-location view, all filterable through Channel, Customer Segment, and Transaction Type Slicers. Every KPI card figure matches the equivalent

Excel, SQL, and Python calculationexactly, and the visual design uses a consistent colour theme and clear titling throughout.

5. Part D: Final Business Insights

Insight 1: Reported Total value overstates real revenue by about 9%

Evidence: All status value is ₦560,477,355; successful only value is ₦510,807,441, comfirmed identically in Excel, SQL, Python, and Power BI.

Business Meaning: ₦49.7 million sits in Reversed, Failed, and Pending transactions. Any report quoting the all-status figure as "revenue" overstates true income by roughly a tenth.

Recommended Action: Standardize all FinTrust reporting on successful only value as the primary KPI, with all-status value retained only as secondary context

Insight 2: The "At Risk" customer segment is both the largest group and the largest pool of value.

Evidence: RFM segmentation places 600 of 1,500 customers (40%) in the At Risk category, holding ₦177,875,451 in successful value, 34.8% of the total, more than any other segment.

Business Meaning: This is not a small pocket of dormant accounts; It is the single largest revenue pool in the business, and it is the one showing clear signs of disengagement.

Recommended Action: Launch a dedicated, targeted re-engagement campaign for the At Risk segment before further value is lost.

Insight 3: Risk concentrates is two transaction types, consistently across every channel.

Evidence: Transfer (24.8-30.6% depending on channel) and Cash Withdrawal (17.0-26.6%) carry the highest risk-review rates across all five channels (ATM, Mobile App, POS, USSD, and Web).

Business Meaning: The elevated risk is a property of the transaction type itself, not a weakness of any single platform or channel.

Recommended Action: Reweight fraud-review staffing and rule-turning toward Transfer and Cash Withdrawal specifically, rather than distributing effort evenly across all transaction types.

Insight 4: Transaction amount is an independent, threshold-driven predictor of risk review.

Evidence: The flag rate holds near 16-19% below ₦100,000, then rises to 25.5% between ₦100,000-250,000, and 40.9% above ₦250,000.

Business Meaning: Large transactions are inappropriately scrutinized, and the relationship is not linear; there is a clear point above which review intensity accelerates sharply..

Recommended Action: Build an explicit high-value review tier in FinTrust's risk policy, rather than relying on type or channel alone to trigger scrutiny.

Insight 5: A small number of customers drive a disappropriate share of revenue.

Evidence: The top 10% of customers by value generate roughly27% of total successful value (confirmed independently in Excel, Python, and SQL); the top 30% generate 59.3%.

Business Meaning: FinTrust's revenue is concentrated, not evenly spread, losing even a handful of top customers would have an outsized effect of total value.

Recommended Action: Establish a dedicated relationship-management programme for the top customer decile, since their retention carries the most financial weight.

6. Part E: Final Validation

Validation 1: Total transaction counts and values.

Validation Method: Independently recomputed from the raw CSVs in Excel, SQL, and Python, then across-checked against Power BI.

Results: 1,500 customers, 12,000 transactions, ₦560,477,355 (all-status) / ₦510,807,441 (successful), identical across all four tools .

Conclusion: Fully validated. No changes are require.

Validation 2: Success rate and risk-review rate.

Validation Method: Recomputed using COUNTIFS in Excel, grouped aggregation in SQL, pandas in Python, and DAX measures in Power BI.

Results: 90.47% success rate and 19.60% risk-review rate; identical across all four tools .

Conclusion: Fully validated. No changes are require.

Validation 3: High-value transaction flag rate.

Validation Method: Re-filtered the raw transaction data directly by amount band (SQL Q7) and recalculated the flag rate for transactions above ₦100,000, checking it against a figure stated in an earlier draft.

Results: The verified rate is 30.0% (transactions above ₦100,000 in Q7's two highest bands). An ealier draft had stated 30.2%.

Conclusion: **This findings changed after validation.** The figure was corrected from 30.2% to 30.0%. The direction and business meaning are unchanged; only the precise figure was corrected, and the corrected figure is now the one used throughout this report.

Validation 4: Customer value concentration.

Validation Method: Tested three independent ways; Excel's Pareto/decile analysis, Python's full-customer-model re-test of the original week 2 estimate, and SQL's NTILE-based decile ranking (Q11).

Results: All three methods agree closely: the top 10% of customers generate roughly a quarter to just over a quarter of total value.

Conclusion: Validated across three independent methods and tools. Original finding confirmed as substantively correct.

Validation 4: Customer value concentration.

Validation Method: Recomputed month-over-month growth independently in Excel, SQL, Python, and Power BI.

Results: February showed a 6.7% decline versus January, followed by an 11.6% recovery in March; identical across all four tools. SQL's per-day analysis (Q8) further shows that daily transaction volume was constant at 133 transactions/day across all three months, meaning the monthly count difference is explained entirely by February having fewer calendar days, not by a genuine drop in daily activity.

Conclusion: Validated and now explained. What initially looked like an unexplained February dip is resolved by Q8: the underlying daily rate never changed, only the number of days in the month did. This finding has moved from "open for investigation" (as it stood in Week 3) to "close and explained" in this final report.

7. Summary

This report closes out the Data Analytics track's contribution to the FinTrust Digital Bank solution. Every headline figure has been independently computed and reconciled across four separate tools; one finding (high-value flag rate) was corrected after re-validation, and one open question from Week 3 (the February dip) has now been fully explained rather than left outstanding. The five business insights above are each grounded in specific, cited evidence and paired with a concrete recommended action, giving FinTrust's management a decision-ready set of findings rather than a descriptive summary of data.